

Research Statement

Agent systems are reconstituting, for the first time outside human society, the political forms modernity dissolved: sovereignty, organic hierarchy, constitutional order. The AI safety community has forecasters, interpretability researchers, and alignment engineers. It does not yet have a political theory of agent societies. I am building one.

The programme is formal, and it already has results. Distributed shutdown is a sovereignty problem — who decides the exception when no principal knows enough to act alone — and it admits information-theoretic lower bounds (AAAI 2027, under review). Encoding AI goals as constitutional principles inherits Gibbard–Satterthwaite impossibility: the first application of social-choice impossibility results to constitutional AI design. The largest published AI-race simulation (~10K runs, ~9.7M trajectory evaluations) shows that reducing hostility *raises* simulated existential risk — Hobbesian selection pressure, measured rather than conjectured. Correct description is the prerequisite for correct intervention; these results are the description.

I build this theory from inside a frontier lab. At ByteDance Seed I own training-data quality for frontier vision-language models — work that carried the model to tool-free SOTA alongside the Gemini series. Political theories of AI are mostly written far from the machinery; frontier engineering mostly runs without political theory. My programme stands on both: production-grade knowledge of how agent systems are actually made, and the conceptual apparatus to say what they are becoming.

A second programme runs in parallel: verification-first AI for science. I build multi-model research pipelines in which no claim survives on eloquence — every result must pass mechanical verification, from SAT certificates to the Lean 4 kernel, before it consumes an expensive model call, and the pipeline never emits *solved*. The same discipline that audits training data audits machine-generated mathematics.

Publications

Under Review — AAAI 2027

Building the Off-Switch: The Three Gates of Pausing a Multi-Principal Agent System

Shuo Li Liu, **Xuanyi Wang**, Haochen Wen.

First complete formal treatment of distributed shutdown authority. Pausing decomposes into three necessary gates — Detection, Attribution, Intervention — and every observed safety failure maps to exactly one. Conditional lower bounds on what each principal must know for a legitimate pause. github.com/Triciaaaaa/three-gates

In the Stream: Agent Memory for the Big World

Xuanyi Wang, Shuo Li Liu, Haochen Wen.

A third path for agent memory beyond weight absorption and retrieval stores: maintain many local models, update all online, act on the best predictor. State of the art on MemoryAgentBench (90.0), above all published systems. github.com/Triciaaaaa/stream-memory

Convergence Is Open: What Are Representations Converging Toward?

Xuanyi Wang, Shuo Li Liu, Haochen Wen.

A gauge-invariance argument showing cross-modal convergence statistics cannot tell world from language — an unrecognized open problem at the center of the platonic representation hypothesis. github.com/Triciaaaaa/convergence-open

Reduced Hostility Increases Simulated Existential Risk Through Race Synchronization

Shuo Li Liu, **Xuanyi Wang**, Haochen Wen.

Largest published AI-race simulation (~10K runs, ~9.7M evaluations). Reducing hostility synchronizes the race and raises existential risk: safety-oriented agents lose the capability frontier. github.com/Triciaaaaa/race-sync-sim

Working Papers

The Formal Limitations of an Artificial Intelligence Constitution

Xuanyi Wang, Shuo Li Liu.

First application of Gibbard–Satterthwaite impossibility to constitutional AI design: aggregated principles inherit conflict from social choice and face underdetermination at once; both failure modes quantified by simulation. Target: AAAI 2027 Alignment Track.

In Progress

Does General Intelligence Specialize?

Pre-registered 1,000-agent LLM swarm; target ICLR 2027. Over \$50K API compute committed. Early findings: enforcement fails silently under illusory selection pressure; collective-welfare behavior emerges unprompted. github.com/Triciaaaaa/swarm-preregistration

Provers Game the Spec: Verifiable Measurement of Specification Gaming in Formal Theorem Proving

Target ICLR 2027. The first per-proof, machine-checkable gaming criterion (replay-transfer) and the first capability-controlled base-vs-RL measurement of specification exploitation in an exact-ground-truth domain (Lean 4). Preregistered hypotheses, kill-gates, and stop-loss rules fixed before execution.

One-Day AI4Science: A 100-Billion-Token Campaign on Forty Open Theory Problems

Forty curated frontiers across economics theory, theoretical physics, and theoretical chemistry, each with a frozen micro-conjecture and a mechanical verifier. Fail-closed: two independent verification paths per claim, clean-room replication, and a pipeline that never emits *solved*. Economics arm executed August 2026; results under expert review. github.com/Triciaaaaa/ai4sci-one-day

The Political Theology of AGI

Research programme and long-form essay: agent societies are a structural recurrence of pre-modern political forms, and safety frameworks misread them by treating agents as instruments rather than participants. Three open problems connecting the formal work above to Schmitt, Hobbes, and social choice. Manuscript in preparation.

Book

Humans, Ethics, Machines (人、伦理、机器人). Contributing author. Peking University Press, 2023.

Grants & Funding

Scaling AI Safety for a Multi-Agent World — Schmidt Sciences, Trustworthy AI Research for Fundamental Safety (Multi-Agent Systems track). \$181,640 over 24 months; named collaborator (PI: Shuo Li Liu). Under review, submitted August 2026.

Education

University of Chinese Academy of Social Sciences — *B.A. in Philosophy*

2021 – 2025

Mathematical logic; logic, language & cognition; political philosophy; applied ethics. Social choice theory and mechanism design self-taught.

Experience

AgentNorm — *Researcher*

2026 – Now

Independent AI-safety research organization (founder: Shuo Li Liu); fiscally sponsored by Ashgro Inc.

- Research home of the multi-agent safety programme: the four AAAI 2027 submissions, the pre-registered SWARM study, and the Schmidt Sciences proposal above are conducted under AgentNorm.

ByteDance Seed — *Data Scientist, VLM Training Data*

Jan 2026 – Now

Frontier foundation-model lab; training data for ByteDance's frontier vision-language models.

- **Data quality governance:** owned quality across the full VLM pipeline (pre-training, SFT, RL); built the error taxonomy and audit framework covering dozens of sources; identified 26.6% problematic datasets and lifted overall data quality by 5pp.
- **Benchmark offensives:** decomposed ZeroBench, BabyVision, and other hard evaluations into targeted sub-benchmarks with matching training sets; model reached tool-free SOTA, level with the Gemini series.
- **Capability closure:** designed the four-level atomic perception taxonomy and the evaluation-to-curation loop; closed gaps in K12 reasoning and video perception; independently led 9-language OCR evaluation end-to-end.

Microsoft Research Asia — *AI Ethics Research Intern*

Jan 2022 – Jul 2022

- Surveyed alignment and interpretability; proposed source-level debiasing for training datasets; contributed to *Humans, Ethics, Machines* (Peking University Press, 2023).

Research Infrastructure

Multi-Agent Race Simulator. Largest published AI-race simulation: ~9.7M trajectory evaluations across ~10K game runs of competitive development between safety-oriented and capability-racing firms. Foundation of the *Reduced Hostility* paper. github.com/Triciaaaaa/race-sync-sim

SWARM Social Simulation. 1,000-agent LLM swarm with citation-based survival, active elimination, and covert adversarial agents; over \$50K API compute committed; pre-registered for ICLR 2027. github.com/Triciaaaaa/swarm-preregistration

math-swarm: Mechanically-Verified Multi-Model Pipeline. Multiple LLM families attack math problems under a hard budget governor; every claim must pass free mechanical verification — SAT + DRAT/LRAT certificates, exhaustive enumeration, the Lean 4 kernel — before consuming an expensive model call. Success criteria hash-frozen before solving starts; independent replication enforced across model families. github.com/Triciaaaaa/math-swarm

Methods

Formal Social choice theory, mechanism design, political philosophy (sovereignty, constitutional design, contract theory), impossibility results, epistemic logic.

AI / ML VLM/LLM training pipelines (pre-training, SFT, RLHF/RLVR), evaluation benchmark design, multi-agent RL, agent memory architectures, representation analysis.

Simulation Large-scale multi-agent social simulation, game-theoretic modeling, Monte Carlo trajectory evaluation.

Languages Mandarin (native), English (fluent). Tools: Python, PyTorch, LangGraph, Claude Code.